

Unifying Special Relativity and Quantum Mechanics

Lecture 19 — Newton's Law from Graviton Exchange, and Why Gravity Must Be Non-Linear

(The spin-2 potential, the inconsistency of linearized gravity,
and the gauging of space-time translations)

Typed version of the handwritten notes, with added explanations in blue boxes

How to read this document

Conventions. Exactly as in Lectures 1–18: normal text and the yellow / pink / grey boxes are a faithful transcription of the handwritten sheets, and the text colours follow the pens of the original (blue, magenta, green, red, pencil). Everything inside a blue box like this one is material I have added; it is *not* in the original notes.

Pagination. These sheets continue the gravity chapter opened in Lecture 16, so the numbering still carries nine stars: [p. 47*****]–[p. 66*****]. Page markers appear in the margin and in the section titles.

One anomaly in this scan. *Overlap at the start.* The first sheet, [p. 47*****], is a repeat: it is the *last* sheet of the Lecture 18 scan. It is transcribed here in full (Section 1) because Lecture 18 broke off exactly in the middle of the spin-2 calculation set up on it. If you compare the two documents you will find that sheet in both; nothing is missing between them.

Two things I flag as they occur. (i) On [p. 47*****] the original writes “if the photon *were massless*” where the formula that follows requires *were massive* — a slip of the pen, corrected and flagged in place. (ii) The notes use a script symbol $\mathcal{H}^{\rho\sigma}(h)$ for the free equation-of-motion operator and $\mathcal{T}^{\rho\sigma}$, $\mathcal{L}^{\mu\nu}$ for two further objects, without ever defining the letters. They are defined here in a blue box when they first appear.

Where we left off — recap of Lecture 18

Lecture 18 fixed the harmonic gauge $\partial_\mu \bar{h}^{\mu\nu} = 0$, counted the two propagating degrees of freedom of the graviton, and computed the graviton propagator in D space-time dimensions,

$$\langle 0|T[h_{\mu\nu}(x)h_{\rho\sigma}(y)]|0\rangle = \int \frac{d^4p}{(2\pi)^4} e^{-ip\cdot(x-y)} \frac{i}{p^2 + i\varepsilon} \frac{1}{2} \left[g_{\mu\rho}g_{\nu\sigma} + g_{\mu\sigma}g_{\nu\rho} - \frac{2}{D-2} g_{\mu\nu}g_{\rho\sigma} \right],$$

which in $D = 4$ is the numerator $\frac{1}{2} [g_{\mu\rho}g_{\nu\sigma} + g_{\mu\sigma}g_{\nu\rho} - g_{\mu\nu}g_{\rho\sigma}]$ used throughout the present lecture.

It then developed the method for extracting a static potential from field theory: put two static classical sources into the Lagrangian, compute the vacuum persistence amplitude $\langle 0|U_I(\infty, -\infty)|0\rangle$, and compare it with e^{-iE_0T} . Applied to the photon this gave the Coulomb potential $E_0 = Q_1Q_2/4\pi r$ — *repulsive* for like charges. Lecture 19 opens by running the identical machine with the graviton, and the answer comes out with the opposite sign.

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PART I — The spin-2 mediated interaction and Newton's law (pp. 47*****–49*****)

1 Recap of the overlapping sheet [p. 47*****]

[p. 47*****]

1) The exchange of a massless photon correctly reproduces the Coulomb potential

- ▶ equal-sign charges repel ($E_0 > 0$)
- ▶ opposite sign charges attract ($E_0 < 0$)

2) If the photon were massless, we would get

$$E_0 = \frac{Q_1 Q_2 e^{-mr}}{4\pi r}$$

short-range potential, with a characteristic range of the order of the Compton wavelength of the particle mediating the interaction.

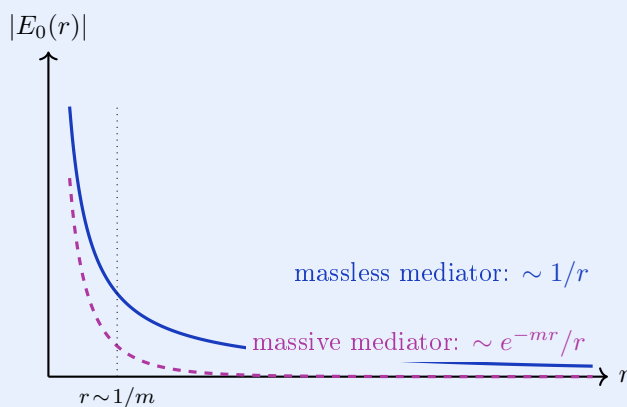
A slip of the pen on this line

The sentence reads “if the photon *were massless*”, but the formula below it contains e^{-mr} , and the previous item has just established what happens when the photon *is* massless. The intended sentence is

*“if the photon were **massive**, we would get . . .”*

This is the Yukawa potential: a mass m in the propagator denominator, $1/(|\vec{p}|^2 + m^2)$ instead of $1/|\vec{p}|^2$, turns the Fourier transform $1/4\pi r$ into $e^{-mr}/4\pi r$. The range is $1/m$, i.e. restoring units the Compton wavelength $\hbar/mc$ of the mediator. Yukawa used exactly this to predict the pion mass from the range of the nuclear force. Photon and graviton are both massless, and that is why both give infinite-range $1/r$ potentials.

The two profiles, side by side



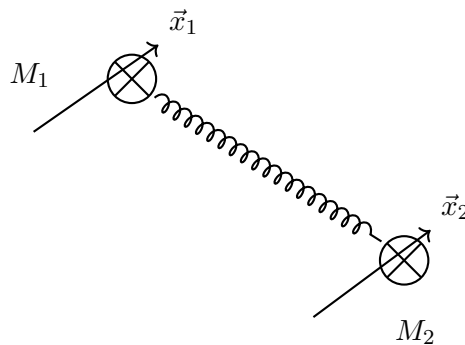
Both curves are $1/r$ at short distance; only the exponential factor distinguishes them, and it does so beyond $r \sim 1/m$. Since gravity is observed to act over astronomical distances, the

mediator must be exactly massless — which is why the whole construction of Lectures 16–18 insisted on a massless spin-2 field.

► The spin-2 mediated interaction

Consider the two sources described by the energy-momentum tensor

$$T_{cl,1}^{\mu\nu}(\vec{x}) = M_1 g^{\mu 0} g^{\nu 0} \delta(\vec{x} - \vec{x}_1), \quad T_{cl,2}^{\mu\nu}(\vec{x}) = M_2 g^{\mu 0} g^{\nu 0} \delta(\vec{x} - \vec{x}_2)$$



They interact with

$$\mathcal{L} = \mathcal{L}_0 - \frac{\kappa}{2} h_{\mu\nu} (T_{cl,1}^{\mu\nu} + T_{cl,2}^{\mu\nu})$$

We use the same coupling κ since it is universal

Reading the two source tensors

$g^{\mu 0} g^{\nu 0}$ is a projector onto the 00 component: it says that the only non-vanishing entry of $T_{cl}^{\mu\nu}$ is T^{00} , the energy density. A static point mass M sitting at $\vec{x}_1$ has energy density $M \delta(\vec{x} - \vec{x}_1)$ and no momentum, no pressure, no stress — exactly what these two lines encode.

The word “universal” in green is the punchline of Lecture 17. The soft-graviton theorem forced every species to couple to $h_{\mu\nu}$ with the *same* κ ; here that result is used, not re-derived. In electromagnetism the analogous statement would be false — different particles carry different charges Q_i — and this is precisely the difference that makes gravity’s “charge” (energy) universal.

2 The computation of the amplitude [p. 48*****]

[p. 48*****]

The computation is similar to the one done in the electromagnetic case.

$$\begin{aligned} \langle 0 | U_I(\infty, -\infty) | 0 \rangle &= 1 - \frac{\kappa^2}{4} \int d^4x d^4y T_{cl,1}^{\mu\nu}(x) T_{cl,2}^{\rho\sigma}(y) \times \\ &\quad \times \underbrace{\langle 0 | T[h_{\mu\nu}(x) h_{\rho\sigma}(y)] | 0 \rangle}_{\text{graviton propagator}} + \dots \end{aligned}$$

$$\begin{aligned}
&= 1 - \frac{\kappa^2}{4} \int d^4x d^4y T_{cl,1}^{\mu\nu}(x) T_{cl,2}^{\rho\sigma}(y) \int \frac{d^4p}{(2\pi)^4} e^{-ip \cdot (x-y)} \frac{i}{p^2 + i\varepsilon} \frac{1}{2} \times \\
&\quad \left[g_{\mu\rho} g_{\nu\sigma} + g_{\mu\sigma} g_{\nu\rho} - g_{\mu\nu} g_{\rho\sigma} \right] \\
&= 1 - \frac{i\kappa^2}{8} \int d^4x d^4y M_1 M_2 g^{\mu 0} g^{\nu 0} g^{\rho 0} g^{\sigma 0} \delta(\vec{x} - \vec{x}_1) \delta(\vec{y} - \vec{x}_2) \int \frac{d^4p}{(2\pi)^4} \frac{e^{-ip \cdot (x-y)}}{p^2 + i\varepsilon} \\
&\quad \times (g_{\mu\rho} g_{\nu\sigma} + g_{\mu\sigma} g_{\nu\rho} - g_{\mu\nu} g_{\rho\sigma}) \\
&= 1 - \frac{i\kappa^2 M_1 M_2}{8} \int dx^0 dy^0 \int \frac{d^4p}{(2\pi)^4} \frac{e^{-ip^0(x^0-y^0) + i\vec{p} \cdot (\vec{x}_1 - \vec{x}_2)}}{(p^0)^2 - |\vec{p}|^2 + i\varepsilon} \times (g_{00}g_{00} + g_{00}g_{00} - g_{00}g_{00}) \\
&= 1 - \frac{i\kappa^2 M_1 M_2}{8} \int dy^0 \int \frac{d^3\vec{p} dp^0}{(2\pi)^3} \frac{\delta(p^0) e^{ip^0 y^0} e^{i\vec{p} \cdot (\vec{x}_1 - \vec{x}_2)}}{(p^0)^2 - |\vec{p}|^2 + i\varepsilon} = \\
&= 1 - \frac{i\kappa^2 M_1 M_2}{8} T \int \frac{d^3\vec{p}}{(2\pi)^3} \frac{e^{i\vec{p} \cdot (\vec{x}_1 - \vec{x}_2)}}{-|\vec{p}|^2 + i\varepsilon}
\end{aligned}$$

The six steps of this calculation, one at a time

This is the single densest page of the lecture, so here is the same chain with the reasons attached.

1. Where the $\kappa^2/4$ comes from. Second-order perturbation theory gives $\frac{(-i)^2}{2!} \int d^4x d^4y \langle T[\mathcal{L}_{int}(x) \mathcal{L}_{int}(y)] \rangle$ with $\mathcal{L}_{int} = -\frac{\kappa}{2} h_{\mu\nu} (T_{cl,1}^{\mu\nu} + T_{cl,2}^{\mu\nu})$. Expanding the square of the sum produces 1–1, 2–2 and twice the cross term 1–2; only the cross term depends on $\vec{x}_1 - \vec{x}_2$ and hence only it contributes to the force. The factor 2 from the cross term cancels the $1/2!$, and $(\kappa/2)^2 = \kappa^2/4$ remains.

2. Insert the propagator. The time-ordered two-point function is exactly what Lecture 18 computed, in $D = 4$.

3. Do the $\vec{x}$ and $\vec{y}$ integrals. The two delta functions set $\vec{x} \rightarrow \vec{x}_1$ and $\vec{y} \rightarrow \vec{x}_2$, which is why $e^{i\vec{p} \cdot (\vec{x}_1 - \vec{x}_2)}$ appears and only the time integrations survive. The i from the propagator numerator combines with $\kappa^2/4 \cdot \frac{1}{2} = \kappa^2/8$.

4. The tensor contraction. With only the 00 components alive,

$$g^{\mu 0} g^{\nu 0} g^{\rho 0} g^{\sigma 0} (g_{\mu\rho} g_{\nu\sigma} + g_{\mu\sigma} g_{\nu\rho} - g_{\mu\nu} g_{\rho\sigma}) = g_{00}g_{00} + g_{00}g_{00} - g_{00}g_{00} = g_{00}g_{00} = 1,$$

since $g_{00} = +1$. The original crosses out the last two terms against each other, which is the same statement.

5. The x^0 integral. $\int dx^0 e^{-ip^0 x^0} = 2\pi \delta(p^0)$; the 2π turns $(2\pi)^4$ into $(2\pi)^3$. This is where “static” enters: no energy flows through the exchanged graviton.

6. The y^0 integral. With p^0 set to zero by the delta function the integrand no longer depends on y^0 , so $\int dy^0 = T$, the (infinite) total time. The denominator becomes $(0)^2 - |\vec{p}|^2 + i\varepsilon = -|\vec{p}|^2 + i\varepsilon$.

The whole amplitude is therefore proportional to T , which is exactly the structure needed for the next step.

3 The result: Newton’s potential [p. 49*****]

We find

[p. 49*****]

$$\langle 0|U_I(\infty, -\infty)|0\rangle = 1 - i \frac{\kappa^2 M_1 M_2}{8} T \int \frac{d^3 \vec{p}}{(2\pi)^3} \frac{e^{i\vec{p} \cdot (\vec{x}_1 - \vec{x}_2)}}{-|\vec{p}|^2 + i\varepsilon}$$

same structure found before

change of sign
(due to the spin of the exchanged particle, since it comes from the propagator numerator)

the massless nature of the graviton gives the same long-range $1/r$ potential

We find

$$E_0 = -\frac{\kappa^2 M_1 M_2}{8} \int \frac{d^3 \vec{p}}{(2\pi)^3} \frac{e^{i\vec{p} \cdot (\vec{x}_1 - \vec{x}_2)}}{|\vec{p}|^2 - i\varepsilon} = -\frac{\kappa^2 M_1 M_2}{32\pi r}$$

↑ the force is always attractive!

If we identify

$$\frac{\kappa^2}{32\pi} = G_N \quad \implies \quad \kappa = \sqrt{32\pi G_N}$$

we get the Newton potential.

From the amplitude to the energy, and the sign

Why E_0 is read off this way. If the two static sources shift the vacuum energy by E_0 , then over a total time T the vacuum simply acquires a phase,

$$\langle 0|U_I(\infty, -\infty)|0\rangle = e^{-iE_0 T} \simeq 1 - iE_0 T + \dots$$

Comparing with the boxed amplitude gives

$$E_0 = \frac{\kappa^2 M_1 M_2}{8} \int \frac{d^3 \vec{p}}{(2\pi)^3} \frac{e^{i\vec{p} \cdot (\vec{x}_1 - \vec{x}_2)}}{-|\vec{p}|^2 + i\varepsilon} = -\frac{\kappa^2 M_1 M_2}{8} \int \frac{d^3 \vec{p}}{(2\pi)^3} \frac{e^{i\vec{p} \cdot (\vec{x}_1 - \vec{x}_2)}}{|\vec{p}|^2 - i\varepsilon},$$

which is the yellow box: the overall minus sign is just -1 pulled out of the denominator.

The integral. With $r = |\vec{x}_1 - \vec{x}_2|$,

$$\int \frac{d^3 \vec{p}}{(2\pi)^3} \frac{e^{i\vec{p} \cdot \vec{r}}}{|\vec{p}|^2} = \frac{1}{(2\pi)^2} \int_0^\infty dp \int_{-1}^1 d\cos\theta e^{ipr\cos\theta} = \frac{1}{2\pi^2 r} \int_0^\infty dp \frac{\sin pr}{p} = \frac{1}{4\pi r},$$

the same integral already done for the photon on [p. 46*****]. Hence $E_0 = -\kappa^2 M_1 M_2 / 32\pi r$.

Why the sign flipped relative to Coulomb. Nothing changed in the method; what changed is the numerator of the propagator. For the photon the contraction of two like currents gave $j^0 j^0 g_{00} = +Q_1 Q_2$ with an overall sign producing $E_0 > 0$. For the graviton the tensor numerator gives $+1$ after the cancellation of step 4 above, but it enters with the opposite overall sign. The general rule, stated in Lecture 18:

spin of mediator	like sources	example
0 (scalar)	attract	Yukawa exchange
1 (vector)	repel	Coulomb, like charges
2 (tensor)	attract	gravity

Each index of the exchanged particle's polarization contributes one factor of (-1) in the contraction of like sources; odd spin therefore repels and even spin attracts.

Why gravity is *always* attractive. M_1 and M_2 are energies, and energies are positive. There is no analogue of “opposite charge”: one cannot make $M_1 M_2 < 0$. So $E_0 < 0$ always. This is the deep reason gravity, alone among the fundamental interactions, never cancels — and why it wins on large scales even though it is by far the weakest.

What $\kappa = \sqrt{32\pi G_N}$ means. Combined with $[\kappa] = [M]^{-1}$ from Lecture 17, this identifies the scale at which the graviton coupling becomes strong:

$$\kappa \sim \frac{1}{M_{\text{Pl}}}, \quad M_{\text{Pl}} = \frac{1}{\sqrt{G_N}} \approx 1.2 \times 10^{19} \text{ GeV}.$$

The programme that began in Lecture 16 with nothing but “take a massless symmetric rank-2 field and forbid ghosts” has now produced, with no geometric input whatsoever: the linearized Einstein equations, the equivalence principle, and Newton's law of gravitation with the correct sign and the correct normalization.

PART II — Why gravity is non-linear (pp. 50*****–58*****)

4 An inconsistency: coupling gravity to a scalar field [p. 50*****]

[p. 50*****]

■ Why is gravity non-linear?

The theory we constructed so far is called “linearized gravity”. It contains correct insights and predictions but it is too simplistic to be effective.

—>► Example: let us show an inconsistency of the theory.

We consider a massless scalar field with Lagrangian

$$\mathcal{L}_\phi = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) \quad \longrightarrow \quad T_\phi^{\mu\nu} = (\partial^\mu \phi)(\partial^\nu \phi) - \frac{1}{2}g^{\mu\nu}(\partial_\rho \phi)(\partial^\rho \phi)$$

The equation of motion is $\square\phi = 0$, and $T_\phi^{\mu\nu}$ is conserved on-shell, since

$$\begin{aligned} \partial_\mu T_\phi^{\mu\nu} &= (\square\phi)(\partial^\nu \phi) + (\partial^\mu \phi)(\partial_\mu \partial^\nu \phi) - \frac{1}{2}\partial^\nu [(\partial_\rho \phi)(\partial^\rho \phi)] = \\ &= (\square\phi)(\partial^\nu \phi) + \cancel{(\partial^\mu \phi)(\partial_\mu \partial^\nu \phi)} - \cancel{(\partial^\nu \partial_\rho \phi)(\partial^\rho \phi)} = \\ &= (\square\phi)(\partial^\nu \phi) = 0 \quad \text{with } \square\phi = 0 \end{aligned}$$

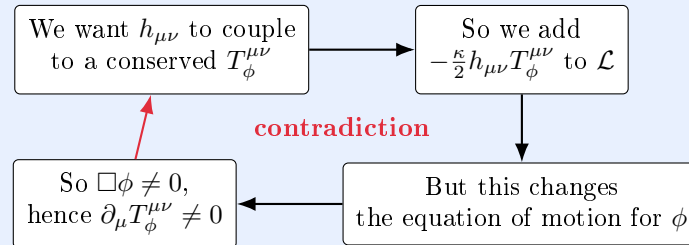
We thus construct the theory

$$\mathcal{L} = \mathcal{L}_0 + \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{\kappa}{2}h_{\mu\nu}T_\phi^{\mu\nu}$$

△ The problem is that $T_\phi^{\mu\nu}$ is no longer conserved since the EOM for ϕ are no longer that of the free theory.

The inconsistency in one line

The logic is a closed loop that fails to close:



Concretely: varying the full $\mathcal{L}$ with respect to ϕ no longer gives $\square\phi = 0$ but

$$\square\phi = \kappa \partial_\mu [h^{\mu\nu} \partial_\nu \phi] + \dots \neq 0,$$

and the computation on this page shows $\partial_\mu T_\phi^{\mu\nu} = (\square\phi)(\partial^\nu \phi)$, which is therefore no longer zero. The very property that justified writing the coupling down — the conservation of $T_\phi^{\mu\nu}$ — is destroyed by writing it down.

Note that the failure is $O(\kappa)$: linearized gravity is fine as a first approximation, and everything computed in Part I (Newton's law) survives. What fails is the attempt to make the theory exact.

5 The physics responsible for the failure [p. 51*****]

[p. 51*****]

The physics responsible for this failure is the following

- i) The gravitational field must couple to the energy-momentum tensor, and the latter must be conserved.
- ii) Considering only $T_\phi^{\mu\nu}$ is obviously insufficient. The gravitational field itself possesses energy and momentum.

↔ It is the total energy-momentum, including that of the gravitational field, that is conserved, not just that stored in the field configuration ϕ

With these considerations in mind, let us consider the case of pure gravity (no additional fields).

■ We start from the free Lagrangian

$$\mathcal{L}_0 = \frac{1}{2}(\partial_\mu h_{\nu\rho})(\partial^\mu h^{\nu\rho}) - (\partial_\mu h_{\nu\rho})(\partial^\nu h^{\mu\rho}) + (\partial_\mu h^{\mu\nu})(\partial_\nu h) - \frac{1}{2}(\partial_\mu h)(\partial^\mu h)$$

↓ From this Lagrangian, we have an energy momentum tensor $T^{\mu\nu}(h)$ which is given by

This is the Fierz–Pauli Lagrangian of Lecture 16

Nothing new is being introduced here: these are the four terms fixed in Lecture 16 by demanding a massless, ghost-free spin-2 field, with $h = h^\mu{}_\mu$ the trace. Reading them left to right:

- $\frac{1}{2}(\partial_\mu h_{\nu\rho})(\partial^\mu h^{\nu\rho})$ — the naive kinetic term, one for each of the 10 components;
- $-(\partial_\mu h_{\nu\rho})(\partial^\nu h^{\mu\rho})$ and $+(\partial_\mu h^{\mu\nu})(\partial_\nu h)$ — the cross terms that make the whole thing gauge invariant under $h_{\mu\nu} \rightarrow h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$;
- $-\frac{1}{2}(\partial_\mu h)(\partial^\mu h)$ — the trace term, whose relative coefficient is what removes the ghost.

Two features matter for what follows. First, $\mathcal{L}_0$ is *quadratic* in h : the free theory is linear, gravitons do not interact. Second, $\mathcal{L}_0$ contains *no* h without a derivative, so $\partial\mathcal{L}_0/\partial h_{\rho\sigma} = 0$ — a fact used silently on [p. 54*****].

6 $T^{\mu\nu}(h)$ and trilinear self-interactions [p. 52***]**

[p. 52*****]

$$T^{\mu\nu}(h) = \frac{\partial\mathcal{L}_0}{\partial(\partial_\mu h_{\rho\sigma})} (\partial^\nu h^{\rho\sigma}) - g^{\mu\nu} \mathcal{L}_0$$

This energy momentum tensor is conserved, $\partial_\mu T^{\mu\nu}(h) = 0$ on the EOM of the free theory

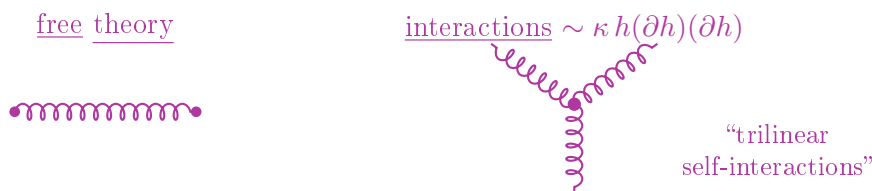
$$\partial_\mu T^{\mu\nu}(h) = 0 \quad \text{with} \quad \mathcal{H}^{\rho\sigma}(h) = 0$$

We do not compute it explicitly, but we know that it has the schematic structure (just counting powers of fields and derivatives)

$$T^{\mu\nu}(h) \sim (\partial h)(\partial h)$$

■ We now move to construct an interacting theory. Following the naive logic, we first write

$$\mathcal{L} = \mathcal{L}_0 - \frac{\kappa}{2} h_{\mu\nu} T^{\mu\nu}(h)$$



As an interacting theory, gravity is forced to have non-linear self-interactions.

What the two boxed formulas actually say

The first box is the canonical (Noether) energy–momentum tensor. For any Lagrangian depending on fields φ and their first derivatives, invariance under $x^\mu \rightarrow x^\mu + a^\mu$

gives the conserved current

$$T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi)} \partial^\nu \varphi - g^{\mu\nu} \mathcal{L}.$$

Here $\varphi = h_{\rho\sigma}$ and $\mathcal{L} = \mathcal{L}_0$. So $T^{\mu\nu}(h)$ is the energy and momentum carried by the *gravitational field itself* — exactly the object point (ii) of the previous page said we could not ignore.

The symbol $\mathcal{H}^{\rho\sigma}(h)$. The notes never define it. It is the free equation-of-motion operator:

$$\mathcal{H}^{\rho\sigma}(h) \equiv \partial_\mu \frac{\partial \mathcal{L}_0}{\partial(\partial_\mu h_{\rho\sigma})} \quad (\text{the free EOM being } \mathcal{H}^{\rho\sigma}(h) = 0),$$

as the underbrace on [p. 54*****] makes explicit. Written out it is the linearized Einstein operator acting on h ; in the harmonic gauge of Lecture 18 it reduces to $\square \bar{h}^{\rho\sigma}$. Its crucial property, used on [p. 55*****], is that it is *identically* divergenceless, $\partial_\rho \mathcal{H}^{\rho\sigma} \equiv 0$, for any h whatsoever — a Noether identity that follows from the gauge invariance of $\mathcal{L}_0$ (the linearized Bianchi identity).

Why $T^{\mu\nu}(h) \sim (\partial h)(\partial h)$. $\mathcal{L}_0$ is quadratic in h with two derivatives, and the Noether formula neither adds nor removes fields or derivatives. So without doing any work we know the answer has two h 's and two ∂ 's. That is all the following argument needs.

Counting powers in $\mathcal{L} = \mathcal{L}_0 - \frac{\kappa}{2} h_{\mu\nu} T^{\mu\nu}(h)$. One h from the explicit factor, two more inside $T^{\mu\nu}(h)$: the interaction is *cubic*. A cubic term in a Lagrangian is a three-point vertex — the graviton emits a graviton. This is the picture on the right of the sketch above, and it is the beginning of the trouble.

7 Self-interactions: gravity versus electromagnetism [p. 53*****]

[p. 53*****]

The presence of “self-interactions” is a crucial difference compared to electromagnetism

→► The photon does not couple to itself. This is because does not have an electromagnetic charge

→► Contrariwise, the graviton has a “gravitational charge” as this is nothing but energy and momentum, and thus it couples to itself

The one-sentence version, and a missing word

The original reads “*this is because does not have an electromagnetic charge*”; the intended sentence is “because **it** does not have an electromagnetic charge”.

The contrast is the heart of the matter:

Photon	Graviton
couples to electric charge	couples to energy–momentum
is itself electrically neutral	itself carries energy–momentum
$\Rightarrow$ does <i>not</i> couple to itself	$\Rightarrow$ <i>must</i> couple to itself
Maxwell theory is linear	gravity cannot be linear

This is not a technical accident; it is forced. Any field with energy sources gravity, and the gravitational field has energy. The same argument explains why the gluon (which carries colour) self-interacts while the photon does not.

■ The Lagrangian $\mathcal{L} = \mathcal{L}_0 - \frac{\kappa}{2} h_{\mu\nu} T^{\mu\nu}(h)$ cannot be the end of the story because it is inconsistent.

△ The argument is the usual one. Since we modified the Lagrangian, the EOM for the field $h_{\mu\nu}$ changed and the tensor $T^{\mu\nu}(h)$ is no longer conserved.

Let us try to be a bit more quantitative on this point.

$$\mathcal{L} = \mathcal{L}_0 - \frac{\kappa}{2} h_{\mu\nu} T^{\mu\nu}(h)$$

We compute the EOM, that are $\frac{\partial \mathcal{L}}{\partial h_{\rho\sigma}} = \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu h_{\rho\sigma})}$

Let's compute the derivatives separately.

8 The “new” EOM and the failure of conservation [p. 54*****]

[p. 54*****]

$$\mathcal{L} = \mathcal{L}_0 - \frac{\kappa}{2} h_{\mu\nu} T^{\mu\nu}$$

$$\frac{\partial \mathcal{L}}{\partial h_{\rho\sigma}} = -\frac{\kappa}{2} T^{\rho\sigma} \qquad \frac{\partial \mathcal{L}}{\partial(\partial_\mu h_{\rho\sigma})} = \frac{\partial \mathcal{L}_0}{\partial(\partial_\mu h_{\rho\sigma})} - \frac{\kappa}{2} h_{\alpha\beta} \frac{\partial T^{\alpha\beta}}{\partial(\partial_\mu h_{\rho\sigma})}$$

Consequently, the EOM are

$$\begin{aligned} \partial_\mu \left[\frac{\partial \mathcal{L}_0}{\partial(\partial_\mu h_{\rho\sigma})} - \frac{\kappa}{2} h_{\alpha\beta} \frac{\partial T^{\alpha\beta}}{\partial(\partial_\mu h_{\rho\sigma})} \right] &= -\frac{\kappa}{2} T^{\rho\sigma} \\ \underbrace{\partial_\mu \frac{\partial \mathcal{L}_0}{\partial(\partial_\mu h_{\rho\sigma})}}_{\text{this is by definition } \mathcal{H}^{\rho\sigma}(h)} - \frac{\kappa}{2} \partial_\mu \left[h_{\alpha\beta} \frac{\partial T^{\alpha\beta}}{\partial(\partial_\mu h_{\rho\sigma})} \right] &= -\frac{\kappa}{2} T^{\rho\sigma} \end{aligned}$$

$$\mathcal{H}^{\rho\sigma} - \frac{\kappa}{2} \partial_\mu \left[h_{\alpha\beta} \frac{\partial T^{\alpha\beta}}{\partial(\partial_\mu h_{\rho\sigma})} \right] = -\frac{\kappa}{2} T^{\rho\sigma}$$

The “new” EOM is thus given by

$$\mathcal{H}^{\rho\sigma} - \frac{\kappa}{2} \partial_\mu \left[h_{\alpha\beta} \frac{\partial T^{\alpha\beta}}{\partial(\partial_\mu h_{\rho\sigma})} \right] = -\frac{\kappa}{2} T^{\rho\sigma} \quad \star$$

It differs from the EOM of the free theory, $\mathcal{H}^{\rho\sigma} = 0$. Conservation of $T^{\rho\sigma}$ is no longer guaranteed.

9 Taking the divergence of ★ [p. 55*****]

[p. 55*****]

We see it immediately from ★ if we take the four-divergence

$$\underbrace{\partial_\rho \mathcal{H}^{\rho\sigma}}_{\substack{=0, \text{ since } \mathcal{H}^{\rho\sigma} \\ \text{is divergenceless}}} - \frac{\kappa}{2} \underbrace{\partial_\rho \partial_\mu \left[h_{\alpha\beta} \frac{\partial T^{\alpha\beta}}{\partial(\partial_\mu h_{\rho\sigma})} \right]}_{\substack{\text{in general, this term} \\ \text{is not equal to zero}}} = -\frac{\kappa}{2} \underbrace{\partial_\rho T^{\rho\sigma}}_{\substack{\partial_\rho T^{\rho\sigma}=0 \text{ is} \\ \text{not guaranteed}}}$$

Why this is fatal, and why it is not surprising

Read the equation from left to right. The first term vanishes *identically* — not because of any equation of motion, but because $\mathcal{H}^{\rho\sigma}$ is built from a gauge-invariant $\mathcal{L}_0$ and therefore obeys the linearized Bianchi identity. The middle term is some complicated expression that has no reason to vanish. Hence the right-hand side, which is $\partial_\rho T^{\rho\sigma}$ up to a constant, is forced to be non-zero.

So the source of the gravitational field is not conserved — and a non-conserved source is not merely inelegant. Coupling $h_{\mu\nu}$ to a non-conserved $T^{\mu\nu}$ breaks gauge invariance, which reintroduces the unphysical polarizations that Lectures 16–18 worked so hard to remove. The theory would propagate ghosts.

Why it is not surprising: the same thing happened one page earlier with the scalar field. There the matter EOM changed; here the graviton's own EOM changes. The pattern is completely general — adding a source term to a Lagrangian always modifies the equations of motion that made the source conserved in the first place.

10 Towards a consistent theory of gravity: the strategy [p. 55***** — p. 56*****]

■ Towards a consistent theory of gravity

Is there a way to solve the above problem? Let us try to setup a strategy

i) We look for a Lagrangian density of the form

$$\mathcal{L} = \mathcal{L}_0 - \frac{\kappa}{2} h_{\mu\nu} \mathcal{L}^{\mu\nu} \quad \blacktriangleright \quad \mathcal{L}^{\mu\nu} \sim (\partial h)(\partial h), \quad \mathcal{L}^{\mu\nu} \neq T^{\mu\nu}(h)$$

ii) This Lagrangian delivers the EOM

[p. 56*****]

$$\begin{aligned} \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu h_{\rho\sigma})} &= \partial_\mu \left[\frac{\partial \mathcal{L}_0}{\partial(\partial_\mu h_{\rho\sigma})} - \frac{\kappa}{2} h_{\alpha\beta} \frac{\partial \mathcal{L}^{\alpha\beta}}{\partial(\partial_\mu h_{\rho\sigma})} \right] = -\frac{\kappa}{2} \mathcal{L}^{\rho\sigma} \\ &= \mathcal{H}^{\rho\sigma} - \frac{\kappa}{2} \partial_\mu \left[h_{\alpha\beta} \frac{\partial \mathcal{L}^{\alpha\beta}}{\partial(\partial_\mu h_{\rho\sigma})} \right] = -\frac{\kappa}{2} \mathcal{L}^{\rho\sigma} \end{aligned}$$

that we write in the form

$$\mathcal{H}^{\rho\sigma} = -\frac{\kappa}{2} \left\{ \mathcal{L}^{\rho\sigma} - \partial_\mu \left[h_{\alpha\beta} \frac{\partial \mathcal{L}^{\alpha\beta}}{\partial(\partial_\mu h_{\rho\sigma})} \right] \right\}$$

iii) We make a simple observation. If we could identify the term in curly brackets as the energy-momentum tensor of the theory, we would have an object that is automatically conserved

$$\mathcal{H}^{\rho\sigma} = -\frac{\kappa}{2} \underbrace{\left\{ \mathcal{L}^{\rho\sigma} - \partial_\mu \left[h_{\alpha\beta} \frac{\partial \mathcal{L}^{\alpha\beta}}{\partial(\partial_\mu h_{\rho\sigma})} \right] \right\}}_{\equiv \mathcal{T}^{\rho\sigma}} = -\frac{\kappa}{2} \mathcal{T}^{\rho\sigma}$$

Since $\partial_\rho \mathcal{H}^{\rho\sigma} = 0 \longrightarrow \partial_\rho \mathcal{T}^{\rho\sigma} = 0$

↔ If we could identify $\mathcal{T}^{\rho\sigma}$ with the energy-momentum tensor, we would have an object automatically conserved

$$\mathcal{T}^{\rho\sigma} \equiv \mathcal{L}^{\rho\sigma} - \partial_\mu \left[h_{\alpha\beta} \frac{\partial \mathcal{L}^{\alpha\beta}}{\partial (\partial_\mu h_{\rho\sigma})} \right], \quad \begin{cases} \partial_\rho \mathcal{T}^{\rho\sigma} = 0 \\ \mathcal{H}^{\rho\sigma} = -\kappa \mathcal{T}^{\rho\sigma} \end{cases}$$

Is this a viable strategy?

The idea behind the strategy — and two notation warnings

The idea. On [p. 52*****] we *chose* the interaction to be $T^{\mu\nu}(h)$ and then *discovered* that the resulting T was not conserved. The new strategy runs the logic backwards: leave the interaction tensor $\mathcal{L}^{\mu\nu}$ *unspecified*, derive the equations of motion, and then *define* the energy-momentum tensor to be whatever combination $\mathcal{T}^{\rho\sigma}$ appears on the right-hand side. Conservation then comes for free, because $\partial_\rho \mathcal{H}^{\rho\sigma} \equiv 0$ is an identity. This is sometimes called the *Noether bootstrap*, or Deser's construction.

Note the price: $\mathcal{T}^{\rho\sigma} \neq \mathcal{L}^{\rho\sigma}$. The tensor we *couple* to and the tensor that is *conserved* are different objects, differing by the total-derivative term. That difference is what the next page turns into an obstruction.

Warning 1 — the symbols. Three script letters are now in play and none is defined in the notes:

$\mathcal{H}^{\rho\sigma}$	free EOM operator, $\partial_\mu \partial \mathcal{L}_0 / \partial (\partial_\mu h_{\rho\sigma})$; identically divergenceless
$\mathcal{L}^{\mu\nu}$	the unknown tensor we couple $h_{\mu\nu}$ to; $\sim (\partial h)(\partial h)$
$\mathcal{T}^{\rho\sigma}$	the combination that is automatically conserved

Warning 2 — a dropped factor of 2. The last magenta line writes $\mathcal{H}^{\rho\sigma} = -\kappa \mathcal{T}^{\rho\sigma}$, while the displayed equation directly above it (and the boxes on [p. 57*****]) have $\mathcal{H}^{\rho\sigma} = -\frac{\kappa}{2} \mathcal{T}^{\rho\sigma}$. The $\frac{1}{2}$ is correct; the shorthand simply drops it. Nothing in the argument depends on the factor.

11 Two obstructions [p. 57*****–p. 58*****]

[p. 57*****]

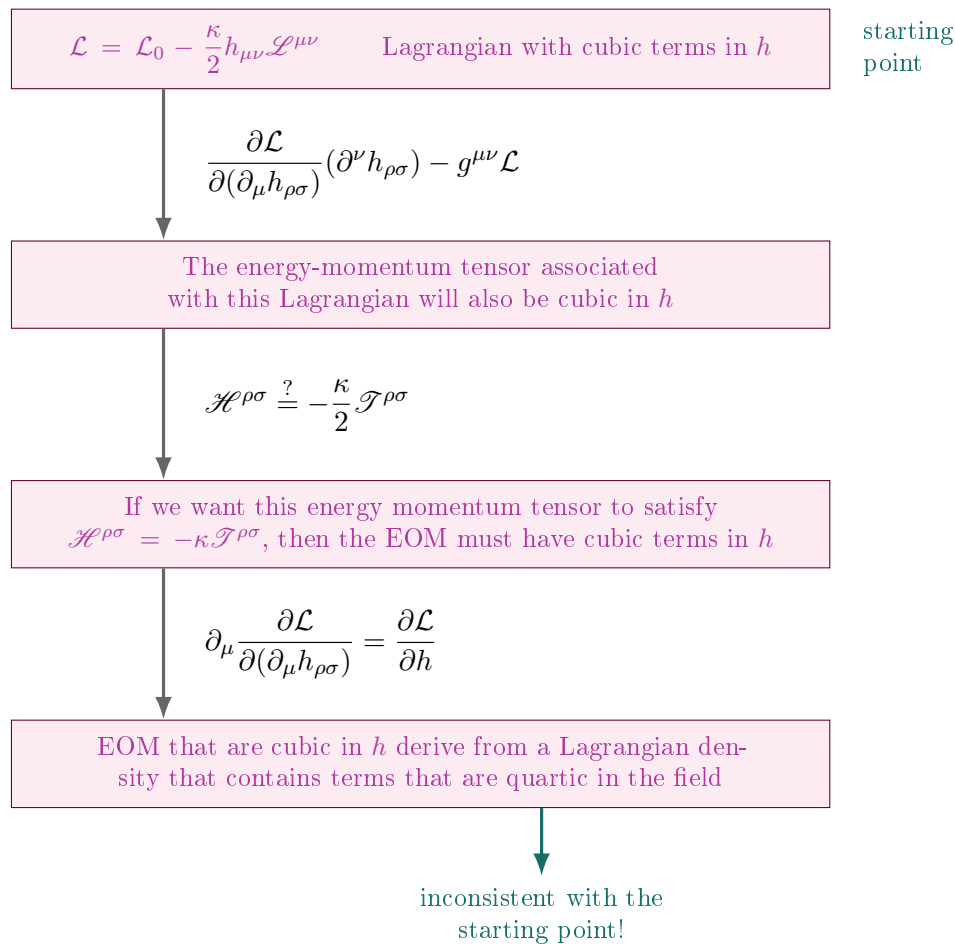
There are two obstructions

① First, we notice that $\mathcal{T}^{\rho\sigma} \neq \mathcal{L}^{\rho\sigma}$. Consequently, the Lagrangian

$$\mathcal{L} = \mathcal{L}_0 - \frac{\kappa}{2} h_{\mu\nu} \mathcal{L}^{\mu\nu}$$

is no longer gauge invariant under the transformation $h_{\mu\nu} \rightarrow h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$

② Second, if we try to derive $\mathcal{T}^{\rho\sigma}$ appearing in $\mathcal{H}^{\rho\sigma} = -\frac{\kappa}{2} \mathcal{T}^{\rho\sigma}$ from Noether theorem starting from the Lagrangian $\mathcal{L} = \mathcal{L}_0 - \frac{\kappa}{2} h_{\mu\nu} \mathcal{L}^{\mu\nu}$, we will find an inconsistency



[p. 58*****]

The counting argument, spelled out

Obstruction 2 is pure bookkeeping in powers of h . Follow one column:

	powers of h in $\mathcal{L}$	powers of h in the EOM
free theory	2 (quadratic)	1 (linear)
our starting point	3 (cubic)	2 (quadratic)
what $\mathcal{H} = -\frac{\kappa}{2} \mathcal{T}$ demands	—	3 (cubic)
... which requires	4 (quartic)	—

Varying a Lagrangian lowers the power of the field by exactly one. So a cubic $\mathcal{L}$ gives quadratic equations of motion. But the consistency condition $\mathcal{H}^{\rho\sigma} = -\frac{\kappa}{2} \mathcal{T}^{\rho\sigma}$ has $\mathcal{T}$ built from the cubic $\mathcal{L}$, hence cubic itself — and cubic equations of motion require a *quartic* Lagrangian. We assumed cubic and derived quartic: contradiction.

Obstruction 1, in one line. $\mathcal{L}_0$ is gauge invariant, and $h_{\mu\nu} \mathcal{L}^{\mu\nu}$ would be too if $\mathcal{L}^{\mu\nu}$ were conserved (the variation $\partial_\mu \xi_\nu \mathcal{L}^{\mu\nu}$ would integrate by parts into $\xi_\nu \partial_\mu \mathcal{L}^{\mu\nu} = 0$). But the conserved object is $\mathcal{T}$, not $\mathcal{L}$, so the variation does not vanish. Gauge invariance is lost at $O(\kappa)$ — which is the second thing the final summary says must be fixed.

This simple chain of consideration implies that our strategy is, at most, an approximation. Schematically

$$\mathcal{L} = \mathcal{L}_0 - \frac{\kappa}{2} h_{\mu\nu} \mathcal{L}^{\mu\nu} + \mathcal{O}(h^4) + \mathcal{O}(h^5) + \dots$$

$$\mathcal{H}^{\rho\sigma} = -\frac{\kappa}{2} \mathcal{T}^{\rho\sigma} + \mathcal{O}(h^3) + \mathcal{O}(h^4) + \dots$$

We thus arrive at a striking conclusion: if we want to construct an interacting theory, we need to:

- ① introduce an infinite series of self-interactions that render the theory fully non-linear
- ② modify the gauge invariant condition $h^{\mu\nu} \rightarrow h^{\mu\nu} + \partial^\mu \xi^\nu + \partial^\nu \xi^\mu$

The bootstrap ladder

The two lines in the pink box are locked together: each order in the Lagrangian generates the next order in the equations of motion, which demands the next order in the Lagrangian, and so on forever.



Nothing is arbitrary along the way: each new term is *determined* by the requirement that the previous one be consistent. The remarkable fact — proved by Deser in 1970, and the content of the rest of this lecture — is that this infinite sum can be resummed in closed form, and that what it sums to is the Einstein–Hilbert action. General relativity is not an extra assumption; it is the unique consistent completion of a massless spin-2 field.

The second requirement is equally telling. The linear gauge symmetry $h \rightarrow h + \partial\xi + \partial\xi$ must be deformed order by order in κ , and the symmetry it deforms into is general coordinate invariance — which is what Part III now constructs.

PART III — General coordinate transformations (pp. 59*****–66*****)

12 The QED template [p. 59*****]

[p. 59*****]

... which is only valid, therefore, in the free-theory limit.

► General coordinate transformations

As we have discussed, going beyond the free theory requires changing the gauge invariance condition

$$h_{\mu\nu} \longrightarrow h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$$

Let us, once more, take inspiration from electromagnetism.
In QED, we did the following:

- 1) We started from a theory with a global $U(1)$ symmetry and gauge invariance for A_μ

$$\begin{cases} \phi(x) \longrightarrow e^{-iq\theta}\phi(x) \\ A_\mu(x) \longrightarrow A_\mu(x) + \partial_\mu\alpha(x) \end{cases}$$

- 2) We “gauged” the global symmetry $U(1)$ that gives the conserved current to which A_μ is coupled:
 $\theta = \theta(x)$
- 3) We set equal the parameters $\alpha(x) = \theta(x)$, defining a “combined” gauge transformation.

The dictionary being set up

The whole of Part III is an exercise in translating one column into the other:

	QED	Gravity
gauge field	A_μ	$h_{\mu\nu}$
conserved current it couples to	j^μ	$T^{\mu\nu}$
global symmetry generating it	$U(1)$ phase	space-time translations
global parameter	θ (constant)	a^μ (constant)
localized parameter	$\theta(x)$	$a^\mu(x)$
gauge parameter of the field	$\alpha(x)$	$\xi^\mu(x)$
identification	$\alpha(x) = \theta(x)$	$\xi^\mu = -\varepsilon^\mu, a^\mu = \kappa\varepsilon^\mu$
resulting local symmetry	local $U(1)$	general coordinate invariance

The one structural difference is that in QED the two symmetries act on *different* fields (θ rotates matter, α shifts A_μ), whereas in gravity both act on $h_{\mu\nu}$ itself — because the graviton is its own source. That is why the final transformation law on [p. 63*****] mixes the two into a single expression with h -dependent terms.

13 Gauging space-time translations [p. 60*****]

[p. 60*****]

To mimic this procedure, we gauge the global symmetry associated to the conserved current to which the field $h_{\mu\nu}$ is coupled

↔ Such global symmetry is given by space-time translations

We implement the strategy as follows

- 1) We start from the free theory $\mathcal{L} = \mathcal{L}_0$ which is:

- i) invariant under a global space-time translation

$$x^\mu \longrightarrow x'^\mu = x^\mu + a^\mu$$

under which the field transform according to

$$h^{\mu\nu}(x) \longrightarrow h'^{\mu\nu}(x) = h^{\mu\nu}(x) - a^\rho \partial_\rho h^{\mu\nu}(x)$$

$$\begin{aligned} x^\mu &\longrightarrow x'^\mu = x^\mu + a^\mu \\ h^{\mu\nu}(x) &\longrightarrow h'^{\mu\nu}(x) = h^{\mu\nu}(x) - a^\rho \partial_\rho h^{\mu\nu}(x) \end{aligned} \quad \star$$

- ii) invariant under a gauge redefinition

$$h^{\mu\nu}(x) \longrightarrow h'^{\mu\nu}(x) = h^{\mu\nu}(x) + \partial^\mu \xi^\nu(x) + \partial^\nu \xi^\mu(x)$$

$$h^{\mu\nu}(x) \longrightarrow h'^{\mu\nu}(x) = h^{\mu\nu}(x) + \partial^\mu \xi^\nu(x) + \partial^\nu \xi^\mu(x) \quad \star\star$$

Where the minus sign in $\star$ comes from

Under $x \rightarrow x' = x + a$ a field is transported, not changed: $h'(x') = h(x)$. Evaluating both sides at the same point x and expanding to first order,

$$h'(x) = h(x' - a) = h(x) - a^\rho \partial_\rho h(x) + O(a^2).$$

That is the whole content of the second line of $\star$: the minus sign is the usual one relating an “active” shift of the field to a “passive” shift of the coordinates. Noether’s theorem applied to this symmetry is what gave $T^{\mu\nu}$ in the first place — which is exactly why translations, and not some other symmetry, are the ones that must be gauged: they are the symmetry whose current is the source of $h_{\mu\nu}$.

14 From global to local translations [p. 61*****]

[p. 61*****]

2) To construct the interaction Lagrangian $\mathcal{L} = \mathcal{L}_0 - \frac{\kappa}{2} h_{\mu\nu} \mathcal{L}^{\mu\nu}$ we “gauge” the global symmetry $\star$, and write

$$x^\mu \rightarrow x'^\mu = x^\mu + a^\mu \quad \xrightarrow[\text{to}]{\text{promoted}} \quad x^\mu \rightarrow x'^\mu = x^\mu + a^\mu(x)$$

↓

global space-time translation

↓

local space-time translation

(a.k.a. infinitesimal general coordinate transformation)

Going from global to local space-time translation also changes the transformation rule of $h_{\mu\nu}(x)$. Let’s work it out.

For an infinitesimal coordinate transformation

$$x^\mu \rightarrow x'^\mu = x^\mu + a^\mu(x)$$

a covariant rank-2 tensor transforms as

$$\begin{aligned} x^\mu &\longrightarrow x'^\mu = x^\mu + a^\mu(x) \\ h_{\mu\nu}(x) &\longrightarrow h'_{\mu\nu}(x') = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} h_{\alpha\beta}(x) \end{aligned}$$

Let us work out this transformation at the first order in “ a ”

15 The transformation law at first order in a [p. 62*****]

[p. 62*****]

$$x^\mu \rightarrow x'^\mu = x^\mu + a^\mu(x)$$

To compute $\frac{\partial x^\alpha}{\partial x'^\mu}$, we need the inverse transformation

$$x^\mu = x'^\mu - a^\mu(x) = x'^\mu - a^\mu(x' - a) = x'^\mu - a^\mu(x') + \mathcal{O}(a^2)$$

From which

$$\frac{\partial x^\alpha}{\partial x'^\mu} = \frac{\partial}{\partial x'^\mu} (x'^\alpha - a^\alpha(x')) = \delta_\mu^\alpha - \partial_\mu a^\alpha$$

We thus have

$$\begin{aligned} h'_{\mu\nu}(x') &= \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} h_{\alpha\beta}(x) = (\delta_\mu^\alpha - \partial_\mu a^\alpha)(\delta_\nu^\beta - \partial_\nu a^\beta) h_{\alpha\beta}(x) = \\ &= h_{\mu\nu}(x) - \delta_\nu^\beta (\partial_\mu a^\alpha) h_{\alpha\beta}(x) - \delta_\mu^\alpha (\partial_\nu a^\beta) h_{\alpha\beta}(x) \\ &= h_{\mu\nu}(x) - (\partial_\mu a^\alpha) h_{\alpha\nu}(x) - (\partial_\nu a^\beta) h_{\mu\beta}(x) \end{aligned}$$

↓ We also write

$$h'_{\mu\nu}(x+a) = h'_{\mu\nu}(x) + a^\rho (\partial_\rho h_{\mu\nu})$$

So that

$$h'_{\mu\nu}(x) = h_{\mu\nu}(x) - a^\rho (\partial_\rho h_{\mu\nu}) - h_{\mu\rho} (\partial_\nu a^\rho) - h_{\rho\nu} (\partial_\mu a^\rho)$$

We thus have, at the first order in “ a ” the transformation

Three terms, three different jobs

The final line has a structure worth naming. It is precisely the *Lie derivative* of $h_{\mu\nu}$ along the vector field a :

$$h'_{\mu\nu}(x) - h_{\mu\nu}(x) = -\mathcal{L}_a h_{\mu\nu} = -\left[a^\rho \partial_\rho h_{\mu\nu} + h_{\mu\rho} \partial_\nu a^\rho + h_{\rho\nu} \partial_\mu a^\rho \right].$$

- $a^\rho \partial_\rho h_{\mu\nu}$ — the *transport* term. It is the only term that survives when a is constant, and it reproduces ★ of [p. 60*****]. Constant a means global translation.
- $h_{\mu\rho} \partial_\nu a^\rho$ and $h_{\rho\nu} \partial_\mu a^\rho$ — the *rotation/stretching* terms. They appear only because a now varies from point to point, i.e. only for a genuinely *local* translation, and they act on the two indices of h one at a time.

Note that both new terms carry an explicit h . This is the technical reason the gauge transformation becomes *field-dependent*, and hence why the symmetry of the interacting theory is not the linear symmetry $h \rightarrow h + \partial\xi + \partial\xi$ of the free theory. It is the deformation that Part II said was unavoidable.

16 The identification $a_\mu = \kappa \varepsilon_\mu$, $\xi_\mu = -\varepsilon_\mu$ [p. 63*****]

[p. 63*****]

$$\begin{aligned} x^\mu &\rightarrow x'^\mu = x^\mu + a^\mu(x) \\ h_{\mu\nu}(x) &\rightarrow h'_{\mu\nu}(x) = h_{\mu\nu}(x) - a^\rho(\partial_\rho h_{\mu\nu}) - h_{\mu\rho}(\partial_\nu a^\rho) - h_{\rho\nu}(\partial_\mu a^\rho) \end{aligned}$$

To which we add the gauge redundancy

$$h_{\mu\nu} \longrightarrow h_{\mu\nu}(x) + \partial_\mu \xi_\nu(x) + \partial_\nu \xi_\mu(x)$$

3) The last step of the strategy requires the identification of $a^\rho(x)$ with the gauge parameter $\xi^\mu(x)$

$\hookrightarrow$ Before making such identification, we note that $a_\mu(x)$ is dimensionfull (with dimension inverse mass) while $\xi_\mu(x)$ is dimensionless.

Since $[\kappa] = [M]^{-1}$, it seems natural to rescale a_μ with κ to make it dimensionless

We set

$$a_\mu(x) \equiv \kappa \varepsilon_\mu(x) ; \quad \xi_\mu(x) = -\varepsilon_\mu(x)$$

The resulting gauge transformation is

$$\begin{aligned} h_{\mu\nu} &\longrightarrow h'_{\mu\nu} = h_{\mu\nu} - \partial_\mu \varepsilon_\nu - \partial_\nu \varepsilon_\mu \\ &\quad - \kappa \varepsilon^\rho(\partial_\rho h_{\mu\nu}) - \kappa h_{\mu\rho}(\partial_\nu \varepsilon^\rho) - \kappa h_{\rho\nu}(\partial_\mu \varepsilon^\rho) \end{aligned}$$



Reading the dimensional argument, and the result

The dimensions. In four dimensions $[\mathcal{L}] = [M]^4$ and $\mathcal{L}_0 \sim (\partial h)^2$, so $[h] = [M]$. The gauge transformation $h \rightarrow h + \partial \xi$ then forces $[\partial \xi] = [M]$, i.e. ξ dimensionless. But a^μ is a *coordinate* shift, $x \rightarrow x + a$, so $[a] = [M]^{-1}$. The two parameters cannot be identified as they stand. Since $[\kappa] = [M]^{-1}$, writing $a_\mu = \kappa \varepsilon_\mu$ makes ε dimensionless and puts both on the same footing.

The minus sign in $\xi_\mu = -\varepsilon_\mu$ is a choice of orientation: it makes the first two terms of the final transformation appear as $-\partial_\mu \varepsilon_\nu - \partial_\nu \varepsilon_\mu$, matching the sign of the Lie-derivative terms so that the whole expression assembles into one object.

What ★ is. Line one is the old linear gauge symmetry; line two is the new, field-dependent piece, suppressed by κ . In the limit $\kappa \rightarrow 0$ (free theory) line two disappears and we recover $h \rightarrow h - \partial \varepsilon - \partial \varepsilon$. So the deformation is controlled by the gravitational coupling, exactly as the bootstrap of Part II predicted: at each order in κ both the Lagrangian and the symmetry get corrected.

17 Observation: $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$ [p. 64*****–p. 65*****]

[p. 64*****]

Observation

The gauge transformation ★ suggests a very neat geometric interpretation

Consider the metric $g_{\mu\nu}$ and promote it to be a function of space-time, $g_{\mu\nu}(x)$, by defining

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + \kappa h_{\mu\nu}(x)$$

where we now indicate with $\eta_{\mu\nu} = (1, -1, -1, -1)$ the flat Minkowski metric

★ Follows from the transformation of the metric tensor under infinitesimal general coordinate transformations

In general, under $x^\mu \rightarrow x'^\mu = x^\mu + \kappa \varepsilon^\mu(x)$

$$g_{\mu\nu}(x) \longrightarrow g'_{\mu\nu}(x') = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} g_{\alpha\beta}(x) \quad \text{and we have}$$

$$\begin{aligned} \eta_{\mu\nu} + \kappa h'_{\mu\nu}(x') &= (\delta_\mu^\alpha - \kappa \partial_\mu \varepsilon^\alpha) (\delta_\nu^\beta - \kappa \partial_\nu \varepsilon^\beta) (\eta_{\alpha\beta} + \kappa h_{\alpha\beta}(x)) \\ &= \eta_{\mu\nu} + \kappa h_{\mu\nu}(x) - \kappa (\partial_\mu \varepsilon^\alpha) \delta_\nu^\beta \eta_{\alpha\beta} - \kappa (\partial_\nu \varepsilon^\beta) \delta_\mu^\alpha \eta_{\alpha\beta} \\ &\quad - \kappa^2 (\partial_\mu \varepsilon^\alpha) \delta_\nu^\beta h_{\alpha\beta} - \kappa^2 (\partial_\nu \varepsilon^\beta) \delta_\mu^\alpha h_{\alpha\beta} = \\ &= \eta_{\mu\nu} + \kappa h_{\mu\nu} - \kappa (\partial_\mu \varepsilon_\nu) - \kappa (\partial_\nu \varepsilon_\mu) \\ &\quad - \kappa^2 (\partial_\mu \varepsilon^\alpha) h_{\alpha\nu} - \kappa^2 (\partial_\nu \varepsilon^\beta) h_{\mu\beta} \end{aligned}$$

[p. 65*****]

Expanding also the left-hand side as

$$\eta_{\mu\nu} + \kappa h'_{\mu\nu}(x') = \eta_{\mu\nu} + \kappa h'_{\mu\nu}(x) + \kappa^2 \varepsilon^\rho (\partial_\rho h_{\mu\nu})$$

We get, from the comparison

$$\begin{aligned} h'_{\mu\nu}(x) &= h_{\mu\nu} - \partial_\mu \varepsilon_\nu - \partial_\nu \varepsilon_\mu - \kappa \varepsilon^\rho (\partial_\rho h_{\mu\nu}) - \kappa (\partial_\mu \varepsilon^\alpha) h_{\alpha\nu} \\ &\quad - \kappa (\partial_\nu \varepsilon^\beta) h_{\mu\beta} \quad \blacksquare \end{aligned}$$

This is the punchline of the entire course so far

Compare the last line with ★ on [p. 63*****]: they are *identical*.

The logic is worth stating in full, because it is easy to lose in the algebra. We never assumed geometry. We started from a massless spin-2 field in flat space, demanded consistency, and were forced to deform the gauge symmetry. Now we discover that the deformed symmetry is nothing other than the statement that

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + \kappa h_{\mu\nu}(x)$$

transforms as a *rank-2 tensor under general coordinate transformations*. In other words: the field-theoretic bootstrap has *derived* the fact that $h_{\mu\nu}$ is a perturbation of the space-time metric, and that the symmetry of the interacting theory is diffeomorphism invariance.

Two remarks on the algebra:

- The factor of κ in $g = \eta + \kappa h$ is what makes the two sides match order by order: η contributes

the $O(\kappa)$ terms $-\kappa\partial\varepsilon$ (giving the *linear* gauge transformation), while h contributes the $O(\kappa^2)$ terms (giving the *field-dependent* piece). Divide through by κ and the two lines of ★ appear automatically.

- The extra term $\kappa^2\varepsilon^\rho\partial_\rho h_{\mu\nu}$ on the left-hand side comes from Taylor expanding $h'(x') = h'(x + \kappa\varepsilon)$; it is what supplies the transport term.

18 The procedure works; the Einstein–Hilbert action [p. 65*****–p. 66*****]

This procedure works, in the following sense.

- Write the Lagrangian

$$\mathcal{L} = \mathcal{L}_0 - \frac{\kappa}{2} h_{\mu\nu} \mathcal{L}^{\mu\nu}, \quad \mathcal{L}^{\mu\nu} \sim (\partial h)(\partial h)$$

taking as $\mathcal{L}^{\mu\nu}$ the most general combination with two derivatives

$$\mathcal{L}^{\mu\nu} = a (\partial^\mu h_{\alpha\beta})(\partial^\nu h^{\alpha\beta}) + b (\partial_\lambda h^{\lambda\mu})(\partial_\alpha h^{\alpha\nu}) + \dots$$

- Imposing invariance under the gauge transformation ★ completely determine all parameters entering in $\mathcal{L}^{\mu\nu}$

[p. 66*****]

- Consequently, one defines the energy-momentum tensor as

$$\mathcal{T}^{\rho\sigma} \equiv \mathcal{L}^{\rho\sigma} - \partial_\mu \left[h_{\alpha\beta} \frac{\partial \mathcal{L}^{\alpha\beta}}{\partial (\partial_\mu h_{\rho\sigma})} \right]$$

In principle, the procedure can be carried out to the next order in powers of κ and so on. Eventually, one reconstruct the full Einstein–Hilbert action

$$S = \frac{1}{16\pi G_N} \int d^4x \sqrt{-g} R(g)$$

invariant under general coordinate transformations and with

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + \kappa h_{\mu\nu}(x)$$

What “the procedure works” means, concretely

The recipe is a fixed-point construction, and each step is finite and mechanical:

1. Write down *every* term of the schematic form $(\partial h)(\partial h)$ with two free indices, each with an unknown coefficient $a, b, c, \dots$ (there are only a handful of independent index contractions).
2. Impose invariance of $\mathcal{L} = \mathcal{L}_0 - \frac{\kappa}{2} h_{\mu\nu} \mathcal{L}^{\mu\nu}$ under ★ to the order in κ being worked at.
3. The requirement fixes *every* coefficient. No freedom is left.
4. Move to the next order in κ and repeat.

The result of doing this to all orders is the expansion of $\frac{1}{16\pi G_N} \sqrt{-g} R(g)$ around flat space with $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$. The Einstein–Hilbert action is not postulated anywhere in this course: it emerges as the unique answer.

Why $\sqrt{-g}$ and R . Once the symmetry is general coordinate invariance, the action must be built from scalars integrated with an invariant measure. $\sqrt{-g} d^4x$ is that measure; $R(g)$, the Ricci scalar, is the unique scalar with exactly two derivatives of the metric. So the *form* of the answer is fixed by the symmetry, and the *normalization* $1/16\pi G_N$ by matching to Newton’s law — which is what Part I computed.

The circle closes. Lecture 16 asked what a consistent massless spin-2 field looks like. Part I of this lecture showed that exchanging it produces Newton’s law with $\kappa^2 = 32\pi G_N$. Part II showed that the theory cannot stop at linear order. Part III showed that the completion is general relativity. Einstein’s route was geometric and started from the equivalence principle; this route is field-theoretic and starts from Lorentz invariance and quantum mechanics. They arrive at the same place.

Summary of Lecture 19 — the logic in nine lines

1. Two static masses coupled as $-\frac{\kappa}{2} h_{\mu\nu} T_{cl}^{\mu\nu}$ exchange one graviton; the vacuum persistence amplitude is $1 - i \frac{\kappa^2 M_1 M_2}{8} T \int \frac{d^3 \vec{p}}{(2\pi)^3} \frac{e^{i\vec{p} \cdot \vec{r}}}{-|\vec{p}|^2 + i\varepsilon}$.
2. Reading it as $e^{-iE_0 T}$ gives $E_0 = -\kappa^2 M_1 M_2 / 32\pi r$: Newton’s potential, with $\kappa^2 = 32\pi G_N$.
3. The sign comes from the propagator *numerator*, i.e. from the spin of the mediator. Even spin attracts, odd spin repels; since masses are positive, gravity is always attractive.
4. Linearized gravity is inconsistent: coupling h to a matter $T_\phi^{\mu\nu}$ changes the matter EOM and destroys $\partial_\mu T_\phi^{\mu\nu} = 0$.
5. The gravitational field itself carries energy, so it must source itself. Coupling h to its own $T^{\mu\nu}(h)$ gives cubic self-interactions.
6. But that has the same disease: the new EOM $\mathcal{H}^{\rho\sigma} - \frac{\kappa}{2} \partial_\mu [\dots] = -\frac{\kappa}{2} T^{\rho\sigma}$ no longer guarantees $\partial_\rho T^{\rho\sigma} = 0$.
7. The bootstrap — define $\mathcal{T}$ as whatever is automatically conserved — hits two obstructions: loss of gauge invariance, and a cubic-EOM/quartic-Lagrangian counting contradiction. Both force an infinite series $\mathcal{O}(h^4) + \mathcal{O}(h^5) + \dots$ and a deformed gauge symmetry.
8. Following the QED template, the symmetry to gauge is space-time translation: $x^\mu \rightarrow x^\mu + a^\mu(x)$, with $a_\mu = \kappa \varepsilon_\mu$ and $\xi_\mu = -\varepsilon_\mu$.
9. The resulting transformation is exactly that of $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$ under infinitesimal general coordinate transformations. Summing the series gives the Einstein–Hilbert action.